

Course Title	VLSI Testing and Validation
Credit Structure	3-0-2-4
Course Code	ED322 / EL464
Program	B.Tech/M.Tech
Category	Core/Technical Elective
Prerequisite courses	Digital Design, Basics of VLSI

Course Objective The course is designed to cover the fundamental aspects of VLSI testing and provide a strong fundamental understanding of test basics: the causes and behavior of manufacturing defects in VLSI technology; defect statistics, fault models; test generation for stuck-at faults, scan design-for-test (DFT), built-in self-test (BIST), memory test, etc. The process of design verification where the basic verification principle and verification methodologies will also be covered. The basic idea is to equip students to envisage testing and verification in a larger design space like SoC.

Suggested Textbooks

1. M.L. Bushnell and V.D. Agrawal, “Essentials of Electronic Testing”, Kluwer Academic Publishers, 2000.
2. L-T. Wang, C-W. Wu and X. Wen, “VLSI Test Principles and Architectures”, Morgan Kaufman Publishers, 2006.

Reference Books

1. H. Fujiwara, “Logic Testing and Design for testability” MIT Press, 1985.
2. N.K. Jha and S. Gupta, “Testing of Digital Systems”, Cambridge University Press, 2004.

Grading

In semester exams- 35%

Final exam – 40%

Project – 25%

Attendance – 70% compulsory to take end semester exam and for project grading

Program Outcomes

At the end of the course student will be able to

- Apply the concepts in testing which can help them design a better yield in IC design.

- Tackle the problems associated with testing of semiconductor circuits at earlier design levels so as to significantly reduce the testing costs.
- Analyze the various test generation methods for static & dynamic CMOS circuits.
- Identify the design for testability methods for combinational & sequential CMOS circuits.
- Recognize the BIST techniques for improving testability.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X		X	X	X							X

Course Content	No. of hours
Introduction to Testing; VLSI Testing Process and Test Equipment; Yield; Reject Rate	4
Fault modeling; Fault equivalence and dominance; fault collapsing; Fault simulation	8
Testability Measures	3
Test generation for combinational circuits	3
Test pattern generation for sequential circuits	3
Memory testing	4
Design for Testability, Built-in self-test techniques	4
Basic Verification Principle, Verification Methodology-Functional verification and simulation based verification, Verification challenges	3
Formal Verification Preliminaries	4
Decision diagram, Equivalence Checking, Model checking	4